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Interactive editing of a machine-generated document

Abstract

Embodiments relate to interactive editing of a machine-generated document. A computer-implemented method includes receiving, by a processor, a machine-generated document and performing a comparison of a current state of the machine-generated document to a previous state. A user edit is identified as one or more user-replaced tokens of a previous token sequence based at least in part on the comparison. A new version of the machine-generated document is generated that includes the one or more user-replaced tokens and identifies one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens. A suggestion list is generated for display to the user in a graphical user interface to indicate the suggested replacement token sequence to replace the one or more related tokens.

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Background/Summary

BACKGROUND

(1) The present invention generally relates to computer systems, and more specifically, to computer-implemented methods, computer systems, and computer program products configured and arranged for interactive editing of a machine-generated document.

(2) A variety of computerized applications, particularly those involving machine intelligence, may generate one or more possible alternative solutions in the course of performing their processing. This is particularly common in applications such as speech recognition, language translation, and other deep learning applications, but can arise with other generative solutions as well. A transformer model can generate natural language text, perform translation between text and natural languages, or translate programs between programming languages. The transformer model typically executes once to produce one or more output products. It is then up to the user to select a preferred output product and make edits to create a final version that is acceptable. Machine generation of documents in general can result in an output that a user may desire to update further for various reasons.

SUMMARY

(3) Embodiments of the present invention are directed to computer-implemented methods for interactive editing of a machine-generated document. A non-limiting example computer-implemented method includes receiving, by a processor, a machine-generated document and performing a comparison of a current state of the machine-generated document to a previous state. A user edit is identified as one or more user-replaced tokens of a previous token sequence based at least in part on the comparison. A new version of the machine-generated document is generated that includes the one or more user-replaced tokens and identifies one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens. A suggestion list is generated for display to the user in a graphical user interface to indicate the suggested replacement token sequence to replace the one or more related tokens.

(4) Other embodiments of the present invention implement features of the above-described method in computer systems and computer program products.

(5) Additional technical features and benefits are realized through the techniques of the present invention. Embodiments and aspects of the invention are described in detail herein and are considered a part of the claimed subject matter. For a better understanding, refer to the detailed description and to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The specifics of the exclusive rights described herein are particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features and advantages of the embodiments of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 depicts a block diagram of an example computing environment for use in conjunction with one or more embodiments of the present invention;

(3) FIG. 2 depicts a block diagram of an example computing environment which is configured to perform interactive editing of a machine-generated document according to one or more embodiments of the present invention;

(4) FIG. 3 is a flowchart of a computer-implemented process for interactive editing of a machine-generated document according to one or more embodiments of the present invention;

(5) FIG. 4 depicts a block diagram of detecting differences using a dynamic programming difference computation according to one or more embodiments of the present invention;

(6) FIG. 5 depicts a block diagram of a displayed region map of regions of agreement and regions

of divergence for a machine-generated document according to one or more embodiments of the present invention;

(7) FIG. 6 depicts a block diagram of a token sequences of a machine-generated document according to one or more embodiments of the present invention;

(8) FIG. 7 depicts a user interface highlighting text in a machine-generated document to be updated by a user according to one or more embodiments of the present invention;

(9) FIG. 8 depicts a user interface highlighting a user-made update to text in a machine-generated document during an editing session according to one or more embodiments of the present invention;

(10) FIG. 9 depicts a user interface presenting a suggestion list of replacement locations of related tokens to be replaced based on a user-made update of a previous token according to one or more embodiments of the present invention; and

(11) FIG. 10 depicts a user interface highlighting accepted suggestions according to one or more embodiments of the present invention.

DETAILED DESCRIPTION

(12) Recently, generative techniques have been applied to the realm of software engineering. Leveraging the naturalness hypothesis, code is a form of human communication with similar statistical properties as natural languages. New progress in neural machine translation (NMT) has demonstrated how unsupervised learning techniques can be used to train models that transform source code from one programming language to another. However, code is unique compared to natural languages. For example, code is much more brittle, and swapping even a few characters or tokens can completely change its meaning or effect. In addition, code demands a certain level of correctness. For example, code either compiles or does not, and it is either correct or contains bugs such as logic errors, security flaws, etc. Code translation output, along with other types of machine-generated documents, may need to be further edited by a user to correct errors in translation or other types of generative output. When edits are made to a machine-generated document, the changes may result in consequences in other portions of the machine-generated document to maintain consistency or correctness. For instance, in the context of code, an edit may impact other portions of code to maintain functionality of the code. However, the impact of such changes may not be readily apparent to the user, particularly where a large corpus is involved.

(13) Embodiments of the invention address the above-described shortcomings by providing computer-implemented methods, computer systems, and computer program products arranged and configured for interactive editing of a machine-generated document. By keeping a generative model involved after one or more edits are made to a machine-generated document, the impact of the edits can be taken into account to generate one or more suggested additional changes to the machine-generated document as a consequence of the edits. For purposes of illustration and explanation, some example scenarios disclose an application of one or more embodiments of the invention in the processing and presentation of results of automatic translation of computer programs from one computer language to another. One or more embodiments provide a graphical user interface in which a user can edit a machine-generated document, such as the output of a translator, to make corrections and updates. Upon making one or more updates through an interactive editing tool, the updates can be identified as user-replaced tokens of previous tokens. Rather than making the user manually determine the impact of the updates to other portions of the machine-generated document, embodiments can trigger automated generation of a new version of the machine-generated document that replaces one or more related tokens with a replacement token associated with the user-replaced token. The user can be prompted to ensure that the suggested updates are acceptable before the machine-generated document is updated to align with the new version of the machine-generated document. Accordingly, in accordance with one or more embodiments, users can make edits to the resulting machine-generated document from a translator and the edits can trigger analysis for other possible related changes that should be made to align

with the edits. A graphical user interface can illustrate/highlight locations in the machine-generated document where the related changes should be made along with the suggested change, according to one or more embodiments of the invention.

(14) Various technical benefits and technical solutions are provided by providing suggested changes to a machine-generated document in response to user-initiated editing. In the context of software development, as noted herein, source code demands a certain level of correctness because code either compiles or does not, is either correct or contains bugs such as logic errors, security flaws, etc., which can cause computer problems when the translated/transcoded source code is compiled or executed by the computer system. When a translator/transcoder converts source code from one programming language like Java, C++, and Python to another for execution by a computer system, there can be differences among the hypotheses or versions of the translated document. One of the versions of the translation, such as the most confident hypothesis/version of the translated document, can be selected for further editing as a machine-generated document for further updating. Where a user identifies that a portion of the machine-generated document should change, for instance, to correct a source code translation error, the change may have a propagation effect that extends beyond simple text matching. Embodiments of the invention are therefore a technical solution to a technical problem. Further, in the context of source code translation, by ensuring that the final version of the translated source code has been analyzed in view of edits in one or more embodiments of the invention, the improved translated source code can prevent or mitigate potential problems when translated code is executed. Accordingly, by novel techniques used in aspects of the invention, the functioning of the computer system itself is improved, as well as the functioning of multiple computer systems interconnected in a cloud environment, all of which can prevent potential exposure to a future or present computer issue.

(15) Various aspects of the present disclosure are described by narrative text, flowcharts, block diagrams of computer systems and/or block diagrams of the machine logic included in computer program product (CPP) embodiments. With respect to any flowcharts, depending upon the technology involved, the operations can be performed in a different order than what is shown in a given flowchart. For example, again depending upon the technology involved, two operations shown in successive flowchart blocks may be performed in reverse order, as a single integrated step, concurrently, or in a manner at least partially overlapping in time.

(16) A computer program product embodiment (“CPP embodiment” or “CPP”) is a term used in the present disclosure to describe any set of one, or more, storage media (also called “mediums”) collectively included in a set of one, or more, storage devices that collectively include machine readable code corresponding to instructions and/or data for performing computer operations specified in a given CPP claim. A “storage device” is any tangible device that can retain and store instructions for use by a computer processor. Without limitation, the computer readable storage medium may be an electronic storage medium, a magnetic storage medium, an optical storage medium, an electromagnetic storage medium, a semiconductor storage medium, a mechanical storage medium, or any suitable combination of the foregoing. Some known types of storage devices that include these mediums include: diskette, hard disk, random access memory (RAM), read-only memory (ROM), erasable programmable read-only memory (EPROM or Flash memory), static random access memory (SRAM), compact disc read-only memory (CD-ROM), digital versatile disk (DVD), memory stick, floppy disk, mechanically encoded device (such as punch cards or pits/lands formed in a major surface of a disc) or any suitable combination of the foregoing. A computer readable storage medium, as that term is used in the present disclosure, is not to be construed as storage in the form of transitory signals per se, such as radio waves or other freely propagating electromagnetic waves, electromagnetic waves propagating through a waveguide, light pulses passing through a fiber optic cable, electrical signals communicated through a wire, and/or other transmission media. As will be understood by those of skill in the art, data is typically moved at some occasional points in time during normal operations of a storage

device, such as during access, de-fragmentation or garbage collection, but this does not render the storage device as transitory because the data is not transitory while it is stored.

(17) Computing environment **100** contains an example of an environment for the execution of at least some of the computer code involved in performing the inventive methods, such as an interactive editing tool **200** (also referred to as block **200**) for a machine-generated document. In addition to block **200**, computing environment **100** includes, for example, computer **101**, wide area network (WAN) **102**, end user device (EUD) **103**, remote server **104**, public cloud **105**, and private cloud **106**. In this embodiment, computer **101** includes processor set **110** (including processing circuitry **120** and cache **121**), communication fabric **111**, volatile memory **112**, persistent storage **113** (including operating system **122** and block **200**, as identified above), peripheral device set **114** (including user interface (UI), device set **123**, storage **124**, and Internet of Things (IoT) sensor set **125**), and network module **115**. Remote server **104** includes remote database **130**. Public cloud **105** includes gateway **140**, cloud orchestration module **141**, host physical machine set **142**, virtual machine set **143**, and container set **144**.

(18) COMPUTER **101** may take the form of a desktop computer, laptop computer, tablet computer, smart phone, smart watch or other wearable computer, mainframe computer, quantum computer or any other form of computer or mobile device now known or to be developed in the future that is capable of running a program, accessing a network or querying a database, such as remote database **130**. As is well understood in the art of computer technology, and depending upon the technology, performance of a computer-implemented method may be distributed among multiple computers and/or between multiple locations. On the other hand, in this presentation of computing environment **100**, detailed discussion is focused on a single computer, specifically computer **101**, to keep the presentation as simple as possible. Computer **101** may be located in a cloud, even though it is not shown in a cloud in FIG. 1. On the other hand, computer **101** is not required to be in a cloud except to any extent as may be affirmatively indicated.

(19) PROCESSOR SET **110** includes one, or more, computer processors of any type now known or to be developed in the future. Processing circuitry **120** may be distributed over multiple packages, for example, multiple, coordinated integrated circuit chips. Processing circuitry **120** may implement multiple processor threads and/or multiple processor cores. Cache **121** is memory that is located in the processor chip package(s) and is typically used for data or code that should be available for rapid access by the threads or cores running on processor set **110**. Cache memories are typically organized into multiple levels depending upon relative proximity to the processing circuitry. Alternatively, some, or all, of the cache for the processor set may be located “off chip.” In some computing environments, processor set **110** may be designed for working with qubits and performing quantum computing.

(20) Computer readable program instructions are typically loaded onto computer **101** to cause a series of operational steps to be performed by processor set **110** of computer **101** and thereby effect a computer-implemented method, such that the instructions thus executed will instantiate the methods specified in flowcharts and/or narrative descriptions of computer-implemented methods included in this document (collectively referred to as “the inventive methods”). These computer readable program instructions are stored in various types of computer readable storage media, such as cache **121** and the other storage media discussed below. The program instructions, and associated data, are accessed by processor set **110** to control and direct performance of the inventive methods. In computing environment **100**, at least some of the instructions for performing the inventive methods may be stored in block **200** in persistent storage **113**.

(21) COMMUNICATION FABRIC **111** is the signal conduction paths that allow the various components of computer **101** to communicate with each other. Typically, this fabric is made of switches and electrically conductive paths, such as the switches and electrically conductive paths that make up busses, bridges, physical input/output ports and the like. Other types of signal communication paths may be used, such as fiber optic communication paths and/or wireless

communication paths.

(22) **VOLATILE MEMORY 112** is any type of volatile memory now known or to be developed in the future. Examples include dynamic type random access memory (RAM) or static type RAM. Typically, the volatile memory is characterized by random access, but this is not required unless affirmatively indicated. In computer **101**, the volatile memory **112** is located in a single package and is internal to computer **101**, but, alternatively or additionally, the volatile memory may be distributed over multiple packages and/or located externally with respect to computer **101**.

(23) **PERSISTENT STORAGE 113** is any form of non-volatile storage for computers that is now known or to be developed in the future. The non-volatility of this storage means that the stored data is maintained regardless of whether power is being supplied to computer **101** and/or directly to persistent storage **113**. Persistent storage **113** may be a read only memory (ROM), but typically at least a portion of the persistent storage allows writing of data, deletion of data and re-writing of data. Some familiar forms of persistent storage include magnetic disks and solid state storage devices. Operating system **122** may take several forms, such as various known proprietary operating systems or open source Portable Operating System Interface type operating systems that employ a kernel. The code included in block **200** typically includes at least some of the computer code involved in performing the inventive methods.

(24) **PERIPHERAL DEVICE SET 114** includes the set of peripheral devices of computer **101**. Data communication connections between the peripheral devices and the other components of computer **101** may be implemented in various ways, such as Bluetooth connections, Near-Field Communication (NFC) connections, connections made by cables (such as universal serial bus (USB) type cables), insertion type connections (for example, secure digital (SD) card), connections made through local area communication networks and even connections made through wide area networks such as the internet. In various embodiments, UI device set **123** may include components such as a display screen, speaker, microphone, wearable devices (such as goggles and smart watches), keyboard, mouse, printer, touchpad, game controllers, and haptic devices. Storage **124** is external storage, such as an external hard drive, or insertable storage, such as an SD card. Storage **124** may be persistent and/or volatile. In some embodiments, storage **124** may take the form of a quantum computing storage device for storing data in the form of qubits. In embodiments where computer **101** is required to have a large amount of storage (for example, where computer **101** locally stores and manages a large database) then this storage may be provided by peripheral storage devices designed for storing very large amounts of data, such as a storage area network (SAN) that is shared by multiple, geographically distributed computers. IoT sensor set **125** is made up of sensors that can be used in Internet of Things applications. For example, one sensor may be a thermometer and another sensor may be a motion detector.

(25) **NETWORK MODULE 115** is the collection of computer software, hardware, and firmware that allows computer **101** to communicate with other computers through WAN **102**. Network module **115** may include hardware, such as modems or Wi-Fi signal transceivers, software for packetizing and/or de-packetizing data for communication network transmission, and/or web browser software for communicating data over the internet. In some embodiments, network control functions and network forwarding functions of network module **115** are performed on the same physical hardware device. In other embodiments (for example, embodiments that utilize software-defined networking (SDN)), the control functions and the forwarding functions of network module **115** are performed on physically separate devices, such that the control functions manage several different network hardware devices. Computer readable program instructions for performing the inventive methods can typically be downloaded to computer **101** from an external computer or external storage device through a network adapter card or network interface included in network module **115**.

(26) **WAN 102** is any wide area network (for example, the internet) capable of communicating computer data over non-local distances by any technology for communicating computer data, now

known or to be developed in the future. In some embodiments, the WAN may be replaced and/or supplemented by local area networks (LANs) designed to communicate data between devices located in a local area, such as a Wi-Fi network. The WAN and/or LANs typically include computer hardware such as copper transmission cables, optical transmission fibers, wireless transmission, routers, firewalls, switches, gateway computers and edge servers.

(27) END USER DEVICE (EUD) **103** is any computer system that is used and controlled by an end user (for example, a customer of an enterprise that operates computer **101**), and may take any of the forms discussed above in connection with computer **101**. EUD **103** typically receives helpful and useful data from the operations of computer **101**. For example, in a hypothetical case where computer **101** is designed to provide a recommendation to an end user, this recommendation would typically be communicated from network module **115** of computer **101** through WAN **102** to EUD **103**. In this way, EUD **103** can display, or otherwise present, the recommendation to an end user. In some embodiments, EUD **103** may be a client device, such as thin client, heavy client, mainframe computer, desktop computer and so on.

(28) REMOTE SERVER **104** is any computer system that serves at least some data and/or functionality to computer **101**. Remote server **104** may be controlled and used by the same entity that operates computer **101**. Remote server **104** represents the machine(s) that collect and store helpful and useful data for use by other computers, such as computer **101**. For example, in a hypothetical case where computer **101** is designed and programmed to provide a recommendation based on historical data, then this historical data may be provided to computer **101** from remote database **130** of remote server **104**.

(29) PUBLIC CLOUD **105** is any computer system available for use by multiple entities that provides on-demand availability of computer system resources and/or other computer capabilities, especially data storage (cloud storage) and computing power, without direct active management by the user. Cloud computing typically leverages sharing of resources to achieve coherence and economies of scale. The direct and active management of the computing resources of public cloud **105** is performed by the computer hardware and/or software of cloud orchestration module **141**. The computing resources provided by public cloud **105** are typically implemented by virtual computing environments that run on various computers making up the computers of host physical machine set **142**, which is the universe of physical computers in and/or available to public cloud **105**. The virtual computing environments (VCEs) typically take the form of virtual machines from virtual machine set **143** and/or containers from container set **144**. It is understood that these VCEs may be stored as images and may be transferred among and between the various physical machine hosts, either as images or after instantiation of the VCE. Cloud orchestration module **141** manages the transfer and storage of images, deploys new instantiations of VCEs and manages active instantiations of VCE deployments. Gateway **140** is the collection of computer software, hardware, and firmware that allows public cloud **105** to communicate through WAN **102**.

(30) Some further explanation of virtualized computing environments (VCEs) will now be provided. VCEs can be stored as “images.” A new active instance of the VCE can be instantiated from the image. Two familiar types of VCEs are virtual machines and containers. A container is a VCE that uses operating-system-level virtualization. This refers to an operating system feature in which the kernel allows the existence of multiple isolated user-space instances, called containers. These isolated user-space instances typically behave as real computers from the point of view of programs running in them. A computer program running on an ordinary operating system can utilize all resources of that computer, such as connected devices, files and folders, network shares, CPU power, and quantifiable hardware capabilities. However, programs running inside a container can only use the contents of the container and devices assigned to the container, a feature which is known as containerization.

(31) PRIVATE CLOUD **106** is similar to public cloud **105**, except that the computing resources are only available for use by a single enterprise. While private cloud **106** is depicted as being in

communication with WAN **102**, in other embodiments a private cloud may be disconnected from the internet entirely and only accessible through a local/private network. A hybrid cloud is a composition of multiple clouds of different types (for example, private, community or public cloud types), often respectively implemented by different vendors. Each of the multiple clouds remains a separate and discrete entity, but the larger hybrid cloud architecture is bound together by standardized or proprietary technology that enables orchestration, management, and/or data/application portability between the multiple constituent clouds. In this embodiment, public cloud **105** and private cloud **106** are both part of a larger hybrid cloud.

(32) FIG. **2** is a block diagram of an example computing environment **201** which is configured to provide interactive editing of a machine-generated document according to one or more embodiments of the inventions. Computing environment **201** can include computer system(s) **202** and computer systems **250**, which may include any of the hardware and software components and functionality discussed in computing environment **100** of FIG. **1**. Similarly, software applications **204** of computer system **202** may include functionality for execution/processing on processor set **110** to operate and function according to one or more embodiments discussed herein, such as interactive editing tool **200**. Additionally, computer system **202** may include one or more processors, such as processor set **110** of FIG. **1**, and memory **208**. Memory **208** can include and/or be representative of any type of storage, system memory, hard disk, etc., discussed herein. Computing environment **201** may be representative of one or more portions of a cloud computing environment. Functions of computing environment **201** can use and/or be implemented in workloads of a workload layer and any of components of a hardware and software layer.

(33) FIG. **3** is a flowchart of a computer-implemented process **300** for interactive editing of a machine-generated document in accordance with one or more embodiments. Computer-implemented process **300** can be executed by computer system **202** in FIG. **2** as further described herein.

(34) At block **302** of computer-implemented process **300**, a software application **204** executing on computer system **202**, such as interactive editing tool **200**, can receive a machine-generated document. The machine-generated document can be generated using the generative model **210** based at least in part on a document **220**. Document **220** may be stored in memory **208**. A software application **204** may receive and/or request document **220** from one of the computer systems **250**. Document **220** can contain data in one format that is to be converted to another, different format that is usable by a computer system. For explanation purposes, document **220** can be a file, such as a computer-executable file, which contains source code in a first programming language that is to be converted to a second program language different from the first programming language, where the first programming language and the second programming language are configured to be executed by a processor. Document **220** is utilized for explanation purposes and ease of understanding, and it should be appreciated that one or more embodiments of the invention are not meant to be limited. In one or more embodiments, document **220** may refer to a general input and/or a request to a generative system. The input to the generative system may include natural language descriptions, data, sensor readings, computer code, a chemical formula, a diagram, natural speech, an image, etc. Additional examples of input to the generative system may include speech recognition (e.g., translation of audio input to text), musical transcription (e.g., translation of audio input to musical notation), natural language translation, and other forms of text generation. Although example scenarios may use the output side as a textual output, it should be recognized that one or more embodiments could have an output of more than a textual output in which differences can be computed over; the output can be segmented into regions where the corresponding regions are established in different alternatives as discussed further herein. The output can include natural language descriptions, data, sensor readings, computer code, a chemical formula, a diagram, natural speech, an image, etc.

(35) The software application **204** can be configured to translate/transcode/generate document **220**

from a first format to a second format using the generative model **210**, and this conversion can result in a single or multiple different machine-generated documents **222_1**, **222_2**, **222_3**, through **222_N**, where N represents the number of the multiple documents. Multiple machine-generated documents **222_1**, **222_2**, **222_3**, through **222_N** can be generally referred to as machine-generated documents **222**. Machine-generated documents **222** can be multiple hypotheses and/or versions of document **220** which has been translated from, for example, the first programming language to the second programming language, where each machine-generated document **222** is different and/or a different version of the output resulting from translating/transcoding the same document **220** or other source material. For explanation purposes, where multiple documents are generated, the software application **204** can determine that machine-generated document **222_1** has the highest confidence in translation to the second programming language, which may use any standard criteria for determining the confidence, such as a confidence score. Accordingly, the other machine-generated documents **222_2** through **222_N** may have a lower confidence. The original document **220** may be a first programming language document for instance, and the multiple machine-generated documents **222** could all be second programming language documents. Multiple machine-generated documents **222** can be different versions of/from each other; lower confidence machine-generated documents **222_2** through **222_N** are different versions of the highest confidence machine-generated document **222_1**. Thus, a machine-generated document **222** can be a translation of a corpus in a language from a first representation to a second representation by a generative model, such as the generative model **210**.

(36) Software application **204** may call and/or employ a standard translation/transcoding program or generative model to translate document **220** from the first programming language to the second programming language. For example, software application **204** may use a transcoding/translation model as the generative model **210**. The generative model **210** can be a trained model that uses machine learning or other approaches to perform document generation. Where the generative model **210** is a transcoding/translation model, the generative model **210** can include the functionality of transcoders as understood by one of ordinary skill in the art, such as but not limited to a transcoder which for unsupervised neural machine translation (NMT) as understood by one of ordinary skill in the art. Although generative model **210** is described in terms of a transcoding/translation model, any type of generative system/model can be used. The generative model **210** can include an algorithm, a rule-based system, and/or any other form of intelligent generative system capable of producing one or more versions of output for consideration.

(37) A machine-generated document, such as machine-generated document **222_1**, can be edited by a user of the interactive editing tool **200** during an editing session. At block **304**, the interactive editing tool **200** can perform or trigger software application **204** to perform a comparison of a current state of the machine-generated document to a previous state. As one example, software application **204** may perform and/or call one or more other software applications to perform a pairwise comparison between the edited version of the machine-generated document **222_1** and a previous version of the machine-generated document **222_1** to determine each of the differences in one or more edited portions of the machine-generated document **222_1**. Software application **204** may use and/or employ standard comparison algorithms, as understood by one of ordinary skill in the art. As depicted in FIG. 4, software application **204** may use and/or employ Myers's algorithm to separately compare the current version of the machine-generated document **222_1** to the previous version of the machine-generated documents **222_1**. As seen in FIG. 4, software application **204** is configured to compute a difference path, which is a series of insertions and deletions that convert the reference version (e.g., previous version of the machine-generated document **222_1**) into the current version. In one or more embodiments, software application **204** can use the Myer's algorithm to compute the least-cost sequence of insertions and deletions to convert the string of characters or tokens across the top to the string of characters or tokens on the left, although other algorithms could be utilized. In the example in FIG. 4, this illustration is

converting the character sequence “ABCABBA” to “CBABAC”. There are many different sequences of insertions and deletions that could achieve this transformation, but if it is established that each insertion and deletion has a constant cost associated with it, then some sequences are more expensive than others. Each sequence is represented by a path through the matrix. For example, one approach could delete all the characters in the first string, and then insert all the characters in the second string. That would be the path that runs along the top and then down the right hand side and would cost **13** assuming a cost of 1 for each insertion or deletion. The least cost path in this case, only requires 3 deletions and 2 insertions, for a cost of 5. Diagonal traversals take advantage of commonality between the source and target strings, resulting in 0 incremental cost. Myer's algorithm computes this least-cost path, as shown in the diagram.

(38) The software application **204** can be configured to determine regions of agreement and regions of divergence based at least in part on the comparison of the previous version of the machine-generated document **222_1** to the current version of the machine-generated document **222_1**, where versions can be tracked in database **232**. Software application **204** can generate a region map **224** of all the regions of agreement and regions of divergence for the machine-generated document **222_1**, as depicted in FIG. 5. Region map **224** is generated and maintained for each unit of the reference version, which is the previous version of the machine-generated document **222_1**. A unit can be a character and/or token in the machine-generated document **222_1**. A character can be an individual letter and/or individual letters used to make a string. A token can be a word, a string a characters (like a variable name), keywords, etc. A token can also be a subword sequence of characters such as a syllable. Any unit for which there is an insertion or deletion in any pairwise comparison with the reference version is part of a region of divergence in region map **224**. Any unit for which there are no insertion or deletions in any of the pairwise comparisons is part of a region of agreement in region map **224**. In some implementations, small regions of agreement can be merged into adjoining regions of divergence, thereby becoming a region of divergence.

(39) At block **306**, the interactive editing tool **200** can perform or trigger software application **204** to identify a user edit as one or more user-replaced tokens of a previous token sequence based at least in part on the comparison. A multiple way difference comparison can be used to track multiple changes to the machine-generated document with divergent regions to identify edits. For instance, an edit can include a sequence of multiple tokens and edits can occur at multiple locations in the machine-generated document **222_1**.

(40) At block **308**, the interactive editing tool **200** can perform or trigger software application **204** to generate a new version of the machine-generated document that includes the one or more user-replaced tokens and identifies one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens. The new version of the machine-generated document can be generated using autoregression to produce a next token based at least in part on one or more previous tokens generated up to a current location within the machine-generated document. For example, generation of the new version of the machine-generated document can be performed incrementally using the generative model **210**. As an alternative, the new version of the machine-generated document can be generated using non-causal generation to search forward and backward from the one or more user-replaced tokens to identify the one or more related tokens to replace with the suggested replacement token sequence. This may use variable length masking of non-edited regions of the document. Retraining of a generative process may be needed for non-causal generation as changes are made.

(41) At block **310**, the interactive editing tool **200** can perform or trigger software application **204** to generate a suggestion list **228** for display to the user in a graphical user interface **206** to indicate the suggested replacement token sequence to replace the one or more related tokens.

(42) At block **312**, the interactive editing tool **200** can perform or trigger software application **204** to confirm user acceptance of replacement of individual portions of the new version of the machine-generated document from the suggestion list **228**. The computer-implemented process **300**

can continue to monitor for further edits after confirming user acceptance of the suggested replacement token sequence at one or more replacement locations in the new version of the machine-generated document and generate further suggested updates of the machine-generated document until the editing session is complete. The interactive editing tool **200** can perform or trigger creation of a new variation of the machine-generated document **222_1** that is subsequently generated by providing the accepted and rejected combinations of tokens to the generative model **210**. For example, accepting suggestions may not lead to using the generative model **210** again, as the suggestions had already been predicted based on user edits. However, if one or more suggestions are rejected, then the non-predicted change can be interpreted as another user edit, which may result in suggested changes elsewhere within the machine-generated document **222_1**. The user rejecting one or more changes from the suggestion list **228** can result in an alternate version of the machine-generated document being selected as the current state.

(43) Software application **204** can be configured to display a region map **224** in conjunction with the machine-generated document **222_1** in a graphical user interface **206**. In one or more embodiments, the user utilizing a mouse, keyboard, etc., can move a pointer or cursor to select (e.g., hover over, right click, etc.) an individual region of divergence from the region map **224** in order to view the display of a suggestion for that particular region in graphical user interface **206**.

(44) FIG. 5 illustrates how adjacent units can be combined by assigning them identical region numbers into individual regions of agreement or divergence, where divergence can represent a user edit or suggested change based on a user edit. In addition to displaying the alternative translation for the unit in a region on the display, software application **204** can be configured to display a link to another machine-generated document, for example, a new version of machine-generated document **222_1**, a portion of the machine-generated document **222_1** in which a suggestion is located, a thumbnail of the machine-generated document **222_1**, and/or an entirety of the machine-generated document **222_1** with the region clearly marked, highlighted, bolded, encircled, etc. As displayed in graphical user interface **206**, software application **204** can be configured to display an option for the user to select the suggestion for the region in the machine-generated document **222_1** to be utilized in place of the region in the current version of the machine-generated document **222_1**. In response to the user selecting the suggested change for the region in the machine-generated document **222_1**, software application **204** can be configured to utilize the suggestion for the region in the machine-generated document **222_1** in a final version document **240** in place of the region in the current version of the machine-generated document **222_1**.

(45) Although only a few regions of divergence are illustrated in region map **224** for explanation purposes, region map **224** could have numerous regions of changes, and the graphical user interface **206** is configured to display the suggestions associated with a user edit for each of the divergent regions when the user selects a divergent region in region map **224** for display of the suggestions. The region map **224** can be used to construct the suggestion list **228**, which is used by graphical user interface **206** to present results to the user, demarcating regions of divergence and presenting alternatives for those regions. Software application **204** can be configured to generate a suggestion list **228** for display to the user in graphical user interface **206**, where suggestion list **228** can be a data structure of suggestions based at least in part on user-replaced tokens.

(46) FIG. 6 depicts display of an example of token sequences **600** of a machine-generated document in accordance with one or more embodiments. A machine-generated document can be generated with a first sequence of tokens **602A**. Tokens **604** in the first sequence of tokens **602A** may be unchanged by a user until reaching tokens **605**. Upon the user making an edit **606**, the first sequence of tokens **602A** can become a previous version of the machine-generated document, and the edited version can be a second sequence of tokens **602B**, which still includes the tokens **604** up to the edit **606** and subsequent tokens after the edit **606**. A new version of the machine-generated document can be generated that includes the one or more user-replaced tokens **608** for the edit **606** and identifies one or more related tokens **607**, **609** to replace with a suggested replacement token

sequence **610**, **612** associated with the one or more user-replaced tokens **608**. A third sequence of tokens **602C** includes the previous tokens from the second sequence of tokens **602B** along with the one or more user-replaced tokens **608** and suggested replacement token sequence **610**, **612**. The current version of machine-generated document displayed to the user still appears based on the second sequence of tokens **602B**. A suggestion list **228** can be generated for display to the user in a graphical user interface to indicate the suggested replacement token sequence **610**, **612** for the user to accept or reject. If the suggested replacement token sequence **610**, **612** is accepted, then the current version of the machine-generated document as displayed is based on the third sequence of tokens **602C** in combination with the tokens **604** up to the region of change. If the user rejects one or more of the suggested replacement token sequence **610**, **612**, the rejection can be deemed a change that may generate a further analysis for new suggestions. For example, if the suggested replacement token sequence **610** is rejected, a fourth sequence of tokens **602D** can include a rejected suggestion **614** and may result in a new suggestion **616** as an alternate suggested replacement token sequence at a subsequent position in the sequence. The rejected suggestion **614** may retain the original value of tokens **607** or can be an alternate user edit made in response to rejecting the suggested replacement token sequence **610**. The new suggestion **616** may be displayed in the suggestion list **228** for the user to accept or reject. If the user accepts the new suggestion **616** then the current version of the machine-generated document as displayed will be updated based on the fourth sequence of tokens **602D** in combination with the previous tokens leading up to the fourth sequence of tokens **602D**. Although only a single edit is depicted in the example of FIG. 6, multiple edits can be performed and analyzed as multiple change regions with corresponding suggestion lists. Accepting or rejecting suggestions in a machine-generated document with multiple edits may trigger an update to subsequent suggestions.

(47) FIG. 7 is a rendering of an example of graphical user interface **206** displaying machine-generated document **222_1**, which can be a highest confidence translation of document **220** from a first programming language to a second programming language or a user-selected translation from among multiple translations. The user may navigate through the machine-generated document **222_1** using the interactive editing tool **200**. Upon recognizing a translation error **702**, such as use of a parameter in a loop structure, the user may edit the machine-generated document **222_1** to correct the translation error **702**. Making a change to the machine-generated document **222_1** during the editing session can result in a state change such that a current state of the machine-generated document **222_1** includes one or more user-replaced tokens that replaced a previous sequence of tokens associated with the translation error **702**. The version of the machine-generated document **222_1** that included the translation error **702** represents a previous state after the change is made. Version history can be tracked, for example, in database **232**. Upon identifying the change, a new version of the machine-generated document **222** can be generated to determine other suggested changes based at least in part on the user-replaced token replacing the previous token associated with the translation error **702**. The current version of the machine-generated document **222_1** without the new translations can continue to be displayed and used in case the user does not accept the suggestions. For example, FIG. 8 depicts a user edit **802** that changes the translation error **702** of “nums” in FIG. 7 to “range (len (nums))” in a looping structure.

(48) FIG. 9 depicts a multiple version difference summary **230** of the machine-generated document **222_1** after performing an updated translation based at least in part on inclusion of the user-replaced token associated with the user edit **802**. The graphical user interface **206** can display artificial intelligence (AI) comments and suggestions **902** with a list of identified user edits and suggested updates as an example of suggestion list **228**. Locations **904** of the suggested updates can be highlighted or otherwise identified to assist the user in understanding the potential impact and scope of the changes. In the example of FIG. 9, changing “nums” to “range (len (nums))” results in a suggestion to change some instances of “i” within the looping structure to “num[i]”. Note that the related tokens for “i” and replacement tokens for “num[i]” can be determined as a result of a

translation with the user edit **802** incorporated and thus is not simply a find and replace operation, as the meaning of the tokens is understood by the generative model **210** to make suggestions that comply with language structure rules.

(49) FIG. **10** depicts the final version **240** of the machine-generated document after user editing and update completing on graphical user interface **206** according to one or more embodiments. In FIG. **10**, accepted suggestions **1002** are highlighted to illustrate that the user opted to accept the suggestions of FIG. **9** at locations **904** to change selected instances of “i” within the looping structure to “num[i]”.

(50) Although the example sequence of FIGS. **7-10** depicts an example of one change in one region of a machine-generated document, it will be understood that multiple edits in multiple regions of the machine-generated document can occur. Further, embodiments can be applied to many domains. For example, in a program translation context, changes to the originally generated translation, such as a data declaration or loop construct a might imply a difference in how data is subsequently referred to or accessed. In a text generation context, changes to the gender of a character could affect pronouns later in the text. Changes to the description of a character may suggest plot alterations. Employing the techniques as described herein in this context may use a low “temperature” (randomness) setting such that the system would regenerate the same text given the same prompt every time and suggestions can be solely due to edits to the prompt, and not to randomness in the generation process. Other uses can include natural language translations, generation of code from natural language descriptions, or any domain where the generated output of a model and the edited version of that output can be represented with a common encoding scheme. In some cases, textual representations of input or output (such as markup language for UIs, or SMILES for molecules) can serve as an intermediate representation that will enable a common encoding. Further, in image processing applications, changes and suggestions can be graphically determined and made. For instance, changing an eye color of one eye in an image can result in a suggestion to make the same change for a corresponding eye of an eye pair in the image.

(51) It is to be understood that although this disclosure includes a detailed description on cloud computing, implementation of the teachings recited herein are not limited to a cloud computing environment. Rather, embodiments of the present invention are capable of being implemented in conjunction with any other type of computing environment now known or later developed.

(52) Various embodiments of the invention are described herein with reference to the related drawings. Alternative embodiments of the invention can be devised without departing from the scope of this invention. Various connections and positional relationships (e.g., over, below, adjacent, etc.) are set forth between elements in the following description and in the drawings. These connections and/or positional relationships, unless specified otherwise, can be direct or indirect, and the present invention is not intended to be limiting in this respect. Accordingly, a coupling of entities can refer to either a direct or an indirect coupling, and a positional relationship between entities can be a direct or indirect positional relationship. Moreover, the various tasks and process steps described herein can be incorporated into a more comprehensive procedure or process having additional steps or functionality not described in detail herein.

(53) One or more of the methods described herein can be implemented with any or a combination of the following technologies, which are each well known in the art: a discrete logic circuit(s) having logic gates for implementing logic functions upon data signals, an application specific integrated circuit (ASIC) having appropriate combinational logic gates, a programmable gate array(s) (PGA), a field programmable gate array (FPGA), etc.

(54) For the sake of brevity, conventional techniques related to making and using aspects of the invention may or may not be described in detail herein. In particular, various aspects of computing systems and specific computer programs to implement the various technical features described herein are well known. Accordingly, in the interest of brevity, many conventional implementation details are only mentioned briefly herein or are omitted entirely without providing the well-known

system and/or process details.

(55) In some embodiments, various functions or acts can take place at a given location and/or in connection with the operation of one or more apparatuses or systems. In some embodiments, a portion of a given function or act can be performed at a first device or location, and the remainder of the function or act can be performed at one or more additional devices or locations.

(56) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(57) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiments were chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(58) The diagrams depicted herein are illustrative. There can be many variations to the diagram or the steps (or operations) described therein without departing from the spirit of the disclosure. For instance, the actions can be performed in a differing order or actions can be added, deleted or modified. Also, the term “coupled” describes having a signal path between two elements and does not imply a direct connection between the elements with no intervening elements/connections therebetween. All of these variations are considered a part of the present disclosure.

(59) The following definitions and abbreviations are to be used for the interpretation of the claims and the specification. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having,” “contains” or “containing,” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a composition, a mixture, process, method, article, or apparatus that comprises a list of elements is not necessarily limited to only those elements but can include other elements not expressly listed or inherent to such composition, mixture, process, method, article, or apparatus.

(60) Additionally, the term “exemplary” is used herein to mean “serving as an example, instance or illustration.” Any embodiment or design described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments or designs. The terms “at least one” and “one or more” are understood to include any integer number greater than or equal to one, i.e. one, two, three, four, etc. The terms “a plurality” are understood to include any integer number greater than or equal to two, i.e. two, three, four, five, etc. The term “connection” can include both an indirect “connection” and a direct “connection.”

(61) The terms “about,” “substantially,” “approximately,” and variations thereof, are intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application. For example, “about” can include a range of $\pm 8\%$ or 5% , or 2% of a given value.

(62) The descriptions of the various embodiments of the present invention have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used

herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments described herein.

Claims

1. A computer-implemented method comprising: receiving, by an interactive editing tool executed by a processor, a machine-generated document that is generated by a generative model, the machine-generated document including a translation of a corpus in a programming language from a first representation to a second representation; performing, by the interactive editing tool executed by the processor, a comparison of a current state of the machine-generated document to a previous state of the machine-generated document, wherein the comparison of the current state of the machine-generated document to the previous state is during an editing session of the machine-generated document through the interactive editing tool; identifying, by the interactive editing tool executed by the processor, a user edit made to the machine-generated document as one or more user-replaced tokens of a previous token sequence in the machine-generated document based at least in part on the comparison; identifying, by the interactive editing tool executed by the processor, one or more related tokens appearing after the one or more user-replaced tokens in the previous token sequence, wherein the one or more related tokens are identified based on the translation including the user edit incorporated using the generative model configured with language structure rules for the translation; generating, by the interactive editing tool executed by the processor, or triggering generation, by an application, of a new version of the machine-generated document that includes the one or more user-replaced tokens and identifies the one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens; generating, by the interactive editing tool executed by the processor, a suggestion list for display to the user in a graphical user interface to indicate the suggested replacement token sequence to replace the one or more related tokens; outputting, by the interactive editing tool executed by the processor, the suggestion list in the graphical user interface during the editing session through the interactive editing tool; confirming, by the interactive editing tool executed by the processor, user acceptance of the suggested replacement token sequence at one or more replacement locations in the new version of the machine-generated document during the editing session; selecting, by the interactive editing tool executed by the processor, an alternate version of the machine-generated document as the current state based at least in part on the user rejecting one or more changes from the suggestion list during the editing session, to make the alternate version of the machine-generated document a basis for subsequent comparisons; and continuing to monitor, by the interactive editing tool executed by the processor, for further edits to the machine-generated document through the graphical user interface after confirming the user acceptance and generating further suggested updates of the machine-generated document until the editing session is complete, wherein the further suggested updates comprise an alternate suggested replacement sequence based on a user rejection of one or more of the further suggested updates during the editing session.
2. The computer-implemented method of claim 1, wherein the new version of the machine-generated document is generated using autoregression to produce a next token based at least in part on one or more previous tokens generated up to a current location within the machine-generated document.
3. The computer-implemented method of claim 1, further comprising performing a multiple way difference comparison to track multiple changes to the machine-generated documents to identify edits.
4. The computer-implemented method of claim 1, wherein the new version of the machine-generated document is generated using non-causal generation to search forward and backward from

the one or more user-replaced tokens to replace the one or more related tokens with the suggested replacement token sequence associated with the one or more user-replaced tokens.

5. A computer system comprising: a memory having computer readable instructions; and one or more processors for executing the computer readable instructions of an interactive editing tool, the computer readable instructions configured to perform operations by the interactive editing tool comprising: receiving a machine-generated document that is generated by a generative model, the machine-generated document including a translation of a corpus in a programming language from a first representation to a second representation; performing a comparison of a current state of the machine-generated document to a previous state of the machine-generated document, wherein the comparison of the current state of the machine-generated document to the previous state is during an editing session of the machine-generated document through the interactive editing tool; identifying a user edit made to the machine-generated document as one or more user-replaced tokens of a previous token sequence in the machine-generated document based at least in part on the comparison; identifying one or more related tokens appearing after the one or more user-replaced tokens in the previous token sequence, wherein the one or more related tokens are identified based on the translation including the user edit incorporated using the generative model configured with language structure rules for the translation; generating or triggering generation, by an application, of a new version of the machine-generated document that includes the one or more user-replaced tokens and identifies the one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens; generating a suggestion list for display to the user in a graphical user interface to indicate the suggested replacement token sequence to replace the one or more related tokens; outputting the suggestion list in the graphical user interface during the editing session through the interactive editing tool; confirming user acceptance of the suggested replacement token sequence at one or more replacement locations in the new version of the machine-generated document during the editing session; selecting an alternate version of the machine-generated document as the current state based at least in part on the user rejecting one or more changes from the suggestion list during the editing session, to make the alternate version of the machine-generated document a basis for subsequent comparisons; and continuing to monitor for further edits to the machine-generated document through the graphical user interface after confirming the user acceptance and generating further suggested updates of the machine-generated document until the editing session is complete, wherein the further suggested updates comprise an alternate suggested replacement sequence based on a user rejection of one or more of the further suggested updates during the editing session.

6. The system of claim 5, wherein the new version of the machine-generated document is generated using autoregression to produce a next token based at least in part on one or more previous tokens generated up to a current location within the machine-generated document.

7. The system of claim 5, wherein the operations further comprise: performing a multiple way difference comparison to track multiple changes to the machine-generated document to identify edits.

8. The system of claim 5, wherein the new version of the machine-generated document is generated using non-causal generation to search forward and backward from the one or more user-replaced tokens to replace the one or more related tokens with the suggested replacement token sequence associated with the one or more user-replaced tokens.

9. A computer program product comprising a computer readable storage medium having program instructions embodied therewith of an interactive editing tool, the program instructions executable by one or more processors and configured to perform operations by the interactive editing tool comprising: receiving a machine-generated document that is generated by a generative model, the machine-generated document including a translation of a corpus in a programming language from a first representation to a second representation; performing a comparison of a current state of the machine-generated document to a previous state of the machine-generated document, wherein the

comparison of the current state of the machine-generated document to the previous state is during an editing session of the machine-generated document through the interactive editing tool; identifying a user edit made to the machine-generated document as one or more user-replaced tokens of a previous token sequence in the machine-generated document based at least in part on the comparison; identifying one or more related tokens appearing after the one or more user-replaced tokens in the previous token sequence, wherein the one or more related tokens are identified based on the translation including the user edit incorporated using the generative model configured with language structure rules for the translation; generating or triggering generation, by an application, of a new version of the machine-generated document that includes the one or more user-replaced tokens and identifies the one or more related tokens to replace with a suggested replacement token sequence associated with the one or more user-replaced tokens; generating a suggestion list for display to the user in a graphical user interface to indicate the suggested replacement token sequence to replace the one or more related tokens; outputting the suggestion list in the graphical user interface during the editing session through the interactive editing tool; confirming user acceptance of the suggested replacement token sequence at one or more replacement locations in the new version of the machine-generated document during the editing session; selecting an alternate version of the machine-generated document as the current state based at least in part on the user rejecting one or more changes from the suggestion list during the editing session, to make the alternate version of the machine-generated document a basis for subsequent comparisons; and continuing to monitor for further edits to the machine-generated document through the graphical user interface after confirming the user acceptance and generating further suggested updates of the machine-generated document until the editing session is complete, wherein the further suggested updates comprise an alternate suggested replacement sequence based on a user rejection of one or more of the further suggested updates during the editing session.

10. The computer program product of claim 9, wherein the operations further comprise: performing a multiple way difference comparison to track multiple versions of the machine-generated document with divergent region alternatives.

11. The computer program product of claim 9, wherein the new version of the machine-generated document is generated using autoregression to produce a next token based at least in part on one or more previous tokens generated up to a current location within the machine-generated document.

12. The computer program product of claim 9, wherein the new version of the machine-generated document is generated using non-causal generation to search forward and backward from the one or more user-replaced tokens to replace the one or more related tokens with the suggested replacement token sequence associated with the one or more user-replaced tokens.
